

WENO Scheme-Based Lattice Boltzmann Flux Solver for Simulation of Compressible Flows

You Li^{1,2}, Hai-Zhuan Yuan^{1,*}, Xiao-Dong Niu², Yu-Yue Yang¹
and Shi Shu¹

¹ School of Mathematics and Computational Science, Xiangtan University,
Hunan 411105, China.

² College of Engineering, Shantou University, Guangdong 515063, China.

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Abstract. In this paper, the finite difference weighed essentially non-oscillatory (WENO) scheme is incorporated into the recently developed lattice Boltzmann flux solver (LBFS) to simulate compressible flows. The resultant WENO-LBFS scheme combines advantages of WENO scheme and LBFS, e.g., high-order accuracy, high resolution and physical robustness. Various numerical tests are carried out to compare the performance of WENO-LBFS and that of other WENO-based flux solvers, including the Lax-Friedrichs flux, the modification of the Harten-Lax-van Leer flux, the multi-stage predictor-corrector flux and the flux limiter centered flux. It turns out that WENO-LBFS is able to capture discontinuous profile in shock waves. Comparatively, the WENO-LBFS scheme costs less CPU time and eases programming.

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1 Introduction

Computational fluid dynamics (CFD) is the product of the modern fluid mechanics, numerical fluid and computer science. It takes the computer as the tool and applies various discrete formats, numerical experiments, computer simulations and analytical studies to solve practical problems. At present, CFD technology is widely used in the civil engineering fields, long-span Bridges, the design of the aircraft and jet engines, water conservancy engineering, etc. The basic equations of computational fluid dynamics can be divided into compressible flow and incompressible flow and can also be divided into the

*Corresponding author. Email address: yhzxtu@163.com (H.-Z. Yuan)

inviscid flow (Euler equations) and viscous flow (Navier-Stokes equations). The compressible flow is widely applied in combustion science, space flight, jet engines, heat transfer, gas turbine, etc. Usually, the shock wave may exist in compressible problems. Therefore, how to develop high-order accuracy and high-resolution numerical schemes is a valuable and challenging research work.

In the past few decades, numerical algorithms for compressible flow have been continuously developed. After von Neumann [17] proposed numerical dissipation format to solve strong intermittent flow problems, Godunov [35] first proposed discontinuous decomposition algorithm in 1959. Godunov algorithm can effectively reflect the physical characteristics of flow, but it only has first order accuracy, ineffective calculation of contact discontinuity, complex programming and intensive computation [11]. Later, van Leer extended Godunov scheme to obtain MUSCL scheme [22]. MUSCL scheme has second order accuracy, but it can not avoid numerical oscillations. Subsequently, Colella and Woodward [32] developed third order Godunov scheme that is called PPM. PPM has a significant improvement in accuracy and resolution, and it can better capture shock waves. In 1973, Discontinuous Galerkin (DG) method was introduced by Reed and Hill [39]. DG method is a kind of non-standard finite element method (FEM), which combines the advantages of finite difference method (FDM) and finite volume method (FVM). DG method can conveniently achieve the adaptive computing, and it can also construct the format of the arbitrary order accuracy [30]. Unfortunately, all these high-order schemes have to introduce a slope limiter to avoid numerical oscillations. In the late 1980s, an essentially non-oscillation (ENO) scheme was proposed by Harten [1]. ENO scheme eases restrictions of non-growing total variation and allows to increase small total variation. What's more, it can achieve consistently high order accuracy, and it can also avoid numerical oscillations. However, it should be mentioned that only the smoothest stencil is adopted to construct ENO scheme and other stencils are abandoned. Basing on that, Liu and Osher et al. [40] proposed a weighted ENO (WENO) scheme using a convex combination of all candidate stencils. WENO scheme has significantly improved in accuracy and resolution comparing with ENO scheme. In the past few decades, finite difference WENO (FD-WENO) scheme [7, 10, 15] and finite volume WENO (FV-WENO) scheme [4, 7, 16, 31] have greatly developed. But Shu [7] strongly suggests using FD-WENO scheme in practice owing that it has high order accuracy, high resolution and a few calculations.

It is generally known that the local Riemann solver is the key of efficient numerical schemes to solve compressible flow problems. First, we briefly review some conventional fluxes to approximate the solution of Riemann problem. The Godunov flux presented by Godunov [35], which is based on the exact Riemann solver, has the smallest numerical viscosity among all monotone fluxes. But the Godunov flux often does not exist explicit formulas to result in a large amount of calculation. The Lax-Friedrichs (LF) flux as an efficient solver to approximate solution of Riemann problem was given by Lax and Friedrichs. It is the most widely used in many numerical schemes for its simplicity. However, the LF flux has a large numerical viscosity which results in the corresponding

numerical schemes with low resolution. The Richtmyer flux [33], which is based on two-step Lax-Wendroff method, has second-order accuracy. The Harten-Lax-van Leer (HLL) flux, which was proposed by Harten [2], is good in capturing shock waves, but it has the poor resolution for contact discontinuities. Later, the HLLC flux [14,18] was presented to overcome the drawback of HLL flux by restoring shear waves and the missing contact, but it involves complex calculations. The first-order centered (FORCE) flux suggested by Toro [13] is the average of the LF flux and the second order Richtmyer flux. Hence, its numerical viscosity is smaller than that of the LF flux. Toro [12] further presented a multi-stage predictor-corrector (MUSTA) flux where the FORCE flux is used as the predictor flux [37]. Also, a flux limiter centered (FLIC) flux, which combines the Richtmyer flux as a high order flux and the FORCE flux as a low order monotone flux, was given by Toro [13,38]. The FLIC flux not only has high order accuracy for the problems with smooth solution but also well done for problems with discontinuous solution. Unfortunately, a limiter has to be introduced to determine the weights of the FORCE flux and the Richtmyer flux. Moreover, Engquist and Osher [3] raised the Engquist-Osher (EO) flux for the scalar case. Then Osher and Solomon extended it to systems, and it was named Osher-Solomon (OS) flux [36]. The OS flux has desirable properties for shock calculations with small numerical viscosity. Obviously, all of the fluxes above are based directly upon characteristic variables. An alternative approach is gas kinetic flux solver which is based on Boltzmann equation [21]. The solver can be well applied to simulate both incompressible and compressible flows [25,28,29,42,43], but it is usually more complicated and less efficient than conventional flux solvers. In addition, Ji et al. [9] proposed a new flux solver called lattice Boltzmann flux solver (LBFS) for simulation of inviscid compressible flows, which is based on the local solution of lattice Boltzmann equation (LBE). In LBFS, the inviscid flux at the cell interface is reconstructed from the local solution of one-dimensional compressible lattice Boltzmann model. LBFS is used to approximate the solution of Riemann problem evaluated by compressible lattice Boltzmann method (LBM) [19,20,24] based on the theory of molecular motion, which has the clear physical background and further improved by Yang et al. [23,24]. Shu et al. [5,6] extended it to simulate incompressible flows. The LBFS can provide good positivity property for simulation of flows with shock waves [24] and can be well applied to simulate both incompressible and compressible flows [5,25,27,44]. However, LBFS only considers the inviscid flow. The non-equilibrium distribution function introduces numerical dissipation for calculation of inviscid flux. To solve viscous flows, the introduced numerical dissipation has to be controlled. Hence, Yang et al. [26] presented a hybrid lattice Boltzmann flux solver for simulation of viscous compressible flows through an introduction of a switch function to make it effectively capture both strong shock waves and thin boundary layers.

Basing on the above reviews, we have known that conventional FD-WENO scheme has high order accuracy, high resolution, robustness and a few calculations to solve compressible flow problems [4,7,16,31], as well as the new flux LBFS based on the theory of molecular motion has good performance to approximate Riemann solver with the clear

physical background, and it has been successfully coupled with MUSCL scheme and PP-MDE scheme [6, 24, 28]. Therefore, combining the advantages of FD-WENO scheme and LBFS is an efficient way to solve compressible flow problems with complex waves including rarefaction waves, shock waves and contact discontinuities and high-frequency waves, etc. Of course, FD-WENO scheme based on lattice Boltzmann flux solver (WENO-LBFS scheme) is destined to have high order accuracy, high resolution and more robustness comparing with WENO scheme based on conventional fluxes (conventional WENO schemes). Recently, it tends to use finite difference method (FDM), finite volume method (FVM) and finite element method (FEM) to solve the discrete velocity Boltzmann equation. In 2007, Wang derived an implicit-explicit finite-difference lattice Boltzmann method (IMEX-FDLBM) [45] for compressible flows. In the IMEX-FDLBM, the implicit-explicit Runge-Kutta scheme is applied for time discretization. In addition, the first-order upwind, second-order TVD scheme and the fifth-order WENO scheme are applied for space discretization. In 2011, Gan researched lattice Boltzmann study on Kelvin-Helmholtz instability [41]. In Gan's work, the forward Euler finite difference scheme is adopted for time discretization, as well as the fifth-order FD-WENO scheme is adopted for space discretization. It should be noted that the object of study is the discrete velocity Boltzmann equation, and WENO scheme is directly used to evaluate numerical fluxes in [41, 45]. Different from previous studies, in our work, Euler equations are the object and FD-WENO scheme is only adopted to reconstruct physical quantities at the left and right sides of cell interface, while LBFS, which is based on non-free parameter D1Q4 lattice Boltzmann model, is used to evaluate numerical fluxes.

The rest of the paper is organized as follows. In Section 2, the descriptions of the WENO-LBFS scheme based on two-dimensional Euler equations are given, where both LBFS and FD-WENO scheme are described along with their basic algorithms. In Section 3, the present scheme is validated by comparing numerical results to exact solutions for two-dimensional Euler equations. Furthermore, we use it to solve one- and two-dimensional compressible flow problems with complex waves to compare with conventional WENO schemes. Finally, the conclusions are given.

2 Numerical scheme

In this paper, the WENO-LBFS scheme, which combines the advantages of WENO scheme and lattice Boltzmann flux solver, is presented. The process of the WENO-LBFS scheme is that WENO scheme is used to reconstruct the flow variables at the sides of cell interface and LBFS is used to evaluate numerical fluxes at cell interface by using flow variables at two sides of interface reconstructed by WENO scheme.

In this section, the descriptions of the WENO-LBFS scheme based on two-dimensional Euler equations are given, where both LBFS and FD-WENO scheme are described along with their basic algorithms.

2.1 Governing equations

In this work, the inviscid compressible flow simulation is considered, the governing equations of which is conventional Euler equations which can be given by

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \quad (2.1)$$

with

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{pmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (\rho E + p)u \end{pmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (\rho E + p)v \end{pmatrix}, \quad (2.2)$$

where ρ and p are the fluid density and pressure field. (u, v) is velocity field. The total energy of fluid E is defined as

$$E = e + \frac{1}{2}(u^2 + v^2). \quad (2.3)$$

Here, $e = p / [(\gamma - 1)\rho]$ is the potential energy of fluid, and the specific heat ratio is $\gamma = 1.4$ for diatomic gas.

2.2 Finite difference scheme

Assume the grid is uniform, Δx is space step in x -direction, and Δy is space step in y -direction. For finite difference scheme, the spatial derivative is discretized by a conservative approximation. Then spatial semi-discretization of Eq. (2.1) can be given by

$$\frac{d\mathbf{U}_{ij}(t)}{dt} = -\frac{1}{\Delta x} (\hat{\mathbf{F}}_{i+\frac{1}{2},j} - \hat{\mathbf{F}}_{i-\frac{1}{2},j}) - \frac{1}{\Delta y} (\hat{\mathbf{G}}_{i,j+\frac{1}{2}} - \hat{\mathbf{G}}_{i,j-\frac{1}{2}}), \quad (2.4)$$

where $\mathbf{U}_{ij}(t)$ is the numerical approximation to the point value $\mathbf{U}(x_i, y_j, t)$, and $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$ are the numerical flux in x -direction and y -direction respectively, which have the following forms

$$\hat{\mathbf{F}}_{i+\frac{1}{2},j} = \hat{\mathbf{F}}(\mathbf{U}_{i-r,j}, \dots, \mathbf{U}_{i+s,j}), \quad (2.5a)$$

$$\hat{\mathbf{G}}_{i,j+\frac{1}{2}} = \hat{\mathbf{G}}(\mathbf{U}_{i,j-r}, \dots, \mathbf{U}_{i,j+s}). \quad (2.5b)$$

Here, multi-variable functions $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$ with $r+s+1 = k$ variables should be Lipschitz continuous and satisfy the following consistent conditions

$$\hat{\mathbf{F}}(\mathbf{U}, \dots, \mathbf{U}) = \mathbf{F}(\mathbf{U}), \quad (2.6a)$$

$$\hat{\mathbf{G}}(\mathbf{U}, \dots, \mathbf{U}) = \mathbf{G}(\mathbf{U}). \quad (2.6b)$$

For convenience, we rewrite the first order ODE system of Eq. (2.4) as compact form

$$\frac{d\mathbf{U}(t)}{dt} = \mathcal{L}(\mathbf{U}). \quad (2.7)$$

Usually, the three-step Runge-Kutta method is applied to the first order ODE system of Eq. (2.7), which is described as [8]

$$\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta t \mathcal{L}(\mathbf{U}^n), \quad (2.8a)$$

$$\mathbf{U}^{(2)} = \frac{3}{4}\mathbf{U}^n + \frac{1}{4}\mathbf{U}^{(1)} + \frac{1}{4}\Delta t \mathcal{L}(\mathbf{U}^{(1)}), \quad (2.8b)$$

$$\mathbf{U}^{n+1} = \frac{1}{3}\mathbf{U}^n + \frac{2}{3}\mathbf{U}^{(2)} + \frac{2}{3}\Delta t \mathcal{L}(\mathbf{U}^{(2)}), \quad (2.8c)$$

where $\mathcal{L}$ is spatial discretization operator, and Δt is time step.

Obviously, in order to obtain a high-accuracy algorithm and make the computation more stable for space discretization, a high-order accuracy, small numerical viscosity and stable numerical flux should be constructed. As mentioned in the introduction, there are kinds of numerical fluxes, such as the Godunov flux [35], the Lax-Friedrichs (LF) flux, the Richtmyer flux [33], the Harten-Lax-van Leer (HLL) flux [2], the HLLC flux [14, 18], the first-order centered (FORCE) flux [13], the multi-stage predictor-corrector (MUSTA) flux [12], the flux limiter centered (FLIC) flux [13, 38], the Engquist-Osher (EO) flux [3] and the Osher-Solomon (OS) flux [36], etc. All of the fluxes above are based directly upon characteristic variables. In our numerical simulations, the LF flux, the HLLC flux, the MUSTA flux and the FLIC flux are chosen to compare with lattice Boltzmann flux solver (LBFS) based on lattice Boltzmann theory proposed by Ji et al. [9]. Next, lattice Boltzmann flux solver for the difference schemes Eq. (2.4) is given.

2.3 Lattice Boltzmann model to evaluate fluxes $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$

Recently, there exists many developed compressible lattice Boltzmann models, such as D1Q4L2 and D1Q5L2 models [19, 20], etc. It should be mentioned that Yang et al. presented a non-free parameter D1Q4 model [24]. The non-free parameter D1Q4 model is applied to simulate compressible flows with a wide range of Mach numbers and faster convergence rate. Next, the non-free parameter D1Q4 lattice Boltzmann model is presented.

2.3.1 Non-free parameter D1Q4 model for compressible flow

The non-free parameter D1Q4 model is proposed by Yang et al. [24], where the equilibrium distribution functions $f_\alpha(\rho, u, p)$ and lattice velocities e_α are given as following

$$f_1(\rho, u, p) = \frac{\rho(-d_1 d_2^2 - d_2^2 u + d_1 u^2 + d_1 c^2 + u^3 + 3uc^2)}{2d_1(d_1^2 - d_2^2)}, \quad (2.9a)$$

$$f_2(\rho, u, p) = \frac{\rho(-d_1 d_2^2 + d_2^2 u + d_1 u^2 + d_1 c^2 - u^3 - 3uc^2)}{2d_1(d_1^2 - d_2^2)}, \quad (2.9b)$$

$$f_3(\rho, u, p) = \frac{\rho(d_1^2 d_2 + d_1^2 u - d_2 u^2 - d_2 c^2 - u^3 - 3uc^2)}{2d_2(d_1^2 - d_2^2)}, \quad (2.9c)$$

$$f_4(\rho, u, p) = \frac{\rho(d_1^2 d_2 - d_1^2 u - d_2 u^2 - d_2 c^2 + u^3 + 3uc^2)}{2d_2(d_1^2 - d_2^2)}, \quad (2.9d)$$

and

$$e_1(\rho, u, p) = d_1, \quad e_2(\rho, u, p) = -d_1, \quad (2.10a)$$

$$e_3(\rho, u, p) = d_2, \quad e_4(\rho, u, p) = -d_2, \quad (2.10b)$$

with

$$d_1 = \sqrt{u^2 + 3c^2 - \sqrt{4u^2 c^2 + 6c^4}}, \quad (2.11a)$$

$$d_2 = \sqrt{u^2 + 3c^2 + \sqrt{4u^2 c^2 + 6c^4}}, \quad (2.11b)$$

and $c = \sqrt{p/\rho}$ is the peculiar velocity of particles.

Furthermore, this model can be used to simulate hypersonic flows with strong waves (see [6, 23, 24] for more details). Consider the following one-dimensional Riemann problem

$$(\rho, u, p) = \begin{cases} (\rho^-, u^-, p^-), & x < 0, \\ (\rho^+, u^+, p^+), & x \geq 0. \end{cases}$$

Then, the equilibrium distribution functions $f_\alpha^R(\rho, u, p)$ at point $x=0$ can be decided by

$$f_1^R = f_1(\rho^-, u^-, p^-), \quad f_2^R = f_2(\rho^+, u^+, p^+), \quad (2.12a)$$

$$f_3^R = f_3(\rho^-, u^-, p^-), \quad f_4^R = f_4(\rho^+, u^+, p^+), \quad (2.12b)$$

and lattice velocities are given by

$$e_1^R = e_1(\rho^-, u^-, p^-), \quad e_2^R = e_2(\rho^+, u^+, p^+), \quad (2.13a)$$

$$e_3^R = e_3(\rho^-, u^-, p^-), \quad e_4^R = e_4(\rho^+, u^+, p^+). \quad (2.13b)$$

Based on the relations between the conserved quantities and the equilibrium density distribution functions, the density ρ^R , velocity u^R and pressure p^R at the point $x=0$ can be evaluated by

$$\rho^R = \sum_{\alpha=1}^4 f_\alpha^R, \quad (2.14a)$$

$$\rho^R u^R = \sum_{\alpha=1}^4 f_\alpha^R e_\alpha^R, \quad (2.14b)$$

$$\left(\frac{1}{\gamma-1} p^R + \frac{1}{2} \rho^R u^R u^R \right) = \sum_{\alpha=1}^4 f_\alpha^R \left(\frac{1}{2} e_\alpha^R e_\alpha^R + \lambda \right), \quad (2.14c)$$

where $\lambda = [1 - (\gamma - 1)/2]e$ is the potential energy of particles.

2.3.2 Non-free parameter D1Q4 model to evaluate numerical flux

Consider the numerical flux $\hat{\mathbf{F}}_{i+\frac{1}{2},j}$ in x -direction at the point $(x_{i+\frac{1}{2}}, y_j)$, where one-dimensional local Riemann problem in x -direction is referred. Based on non-free parameter D1Q4 model, the density ρ^R , velocity component u^R , and pressure p^R can be evaluated by Eqs. (2.14). However, another velocity component v^R should be evaluated, which is given by

$$\rho^R v^R = \sum_{\alpha=1,3} f_{\alpha}^R \cdot v^{-} + \sum_{\alpha=2,4} f_{\alpha}^R \cdot v^{+}. \quad (2.15)$$

Thus, the numerical flux $\hat{\mathbf{F}}_{i+\frac{1}{2},j}$ can be evaluated as following

$$\hat{\mathbf{F}} = \begin{pmatrix} \hat{F}_1 \\ \hat{F}_2 \\ \hat{F}_3 \\ \hat{F}_4 \end{pmatrix} = \begin{pmatrix} \sum_{\alpha=1}^4 f_{\alpha}^R e_{\alpha}^R \\ \sum_{\alpha=1}^4 f_{\alpha}^R e_{\alpha}^R e_{\alpha}^R \\ \hat{F}_1 v^R \\ \sum_{\alpha=1}^4 f_{\alpha}^R e_{\alpha}^R \left(\frac{1}{2} e_{\alpha}^R e_{\alpha}^R + \lambda \right) + \frac{1}{2} \hat{F}_1 v^R v^R \end{pmatrix}, \quad (2.16)$$

where f_{α}^R and e_{α}^R are computed by Eqs. (2.12) and (2.13).

Similarly with $\hat{\mathbf{F}}_{i+\frac{1}{2},j}$, the numerical flux $\hat{\mathbf{G}}_{i,j+\frac{1}{2}}$ in y -direction at point $(x_i, y_{j+\frac{1}{2}})$ can be evaluated by

$$\hat{\mathbf{G}} = \begin{pmatrix} \hat{G}_1 \\ \hat{G}_2 \\ \hat{G}_3 \\ \hat{G}_4 \end{pmatrix} = \begin{pmatrix} \sum_{\alpha=1}^4 f_{\alpha}^R e_{\alpha}^R \\ \hat{G}_1 u^R \\ \sum_{\alpha=1}^4 f_{\alpha}^R e_{\alpha}^R e_{\alpha}^R \\ \sum_{\alpha=1}^4 f_{\alpha}^R e_{\alpha}^R \left(\frac{1}{2} e_{\alpha}^R e_{\alpha}^R + \lambda \right) + \frac{1}{2} \hat{G}_1 u^R u^R \end{pmatrix}. \quad (2.17)$$

Here, f_{α}^R and e_{α}^R are computed by Eqs. (2.12) and (2.13), where $u^{\pm}$ are replaced by $v^{\pm}$, and u^R is given by

$$\rho^R u^R = \sum_{\alpha=1,3} f_{\alpha}^R \cdot u^{-} + \sum_{\alpha=2,4} f_{\alpha}^R \cdot u^{+}. \quad (2.18)$$

Obviously, the key to obtain high-order and robust numerical fluxes $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$ given by Eqs. (2.16) and (2.17) is how to reconstruct the values (including $\rho^{\pm}$, $u^{\pm}$, $v^{\pm}$ and $p^{\pm}$) with high-order accuracy and high resolution. As mentioned in introduction, there are kinds of reconstruction schemes, such as MUSCL scheme [22], PPM [32], ENO scheme [1] and WENO scheme [7, 15, 40]. Given that, WENO scheme is not only a high-order reconstruction scheme but also can avoid numerical oscillations very well, so we adopt it to reconstruct the values $\rho^{\pm}$, $u^{\pm}$, $v^{\pm}$ and $p^{\pm}$.

2.4 WENO scheme to reconstruct the values $\rho^\pm, u^\pm, v^\pm$ and $p^\pm$

Consider the reconstruction of the flow variables $\rho^\pm, u^\pm, v^\pm$ and $p^\pm$ in x -direction at the point $(x_{i+\frac{1}{2}}, y_j)$. Usually, the conservation variables $\mathbf{U}^\pm$ are reconstructed in physical space directly, and then the flow variables can be computed. Unfortunately, the reconstruction process can not deal with the shock waves problems very well, as the discontinuous solution with low resolution and numerical oscillations. Owing to that, we adopt another way proposed by Shu [7], where the reconstruction process is done in characteristic space. Next, the details for the whole reconstruction process are given.

As the conservation variables $\mathbf{U}$ are known for all grid nodes (x_i, y_j) , we have the local left eigenvectors L^F and right eigenvectors R^F of the Jacobian matrix $\frac{\partial \mathbf{F}}{\partial \mathbf{U}}|_{x_{i+\frac{1}{2}}, y_j}$ as following

$$R^F = \begin{pmatrix} 1 & 1 & 1 & 1 \\ u-c & u & u & u+c \\ v & 0 & 1 & v \\ \gamma e + \frac{u^2+v^2}{2} - uc & \frac{u^2-v^2}{2} & v + \frac{u^2-v^2}{2} & \gamma e + \frac{u^2+v^2}{2} + uc \end{pmatrix}, \quad (2.19a)$$

$$L^F = \begin{pmatrix} \frac{u^2+v^2}{4\gamma e} + \frac{u}{2c} & -\frac{1}{2c} - \frac{u}{2\gamma e} & -\frac{v}{2\gamma e} & \frac{1}{2\gamma e} \\ 1 - \frac{u^2+v^2}{2\gamma e} + \frac{v(u^2+v^2)}{2\gamma e} & \frac{u-uv}{\gamma e} & \frac{v-v^2}{\gamma e} - 1 & \frac{v-1}{\gamma e} \\ -\frac{v(u^2+v^2)}{2\gamma e} & \frac{uv}{\gamma e} & 1 + \frac{v^2}{\gamma e} & -\frac{v}{\gamma e} \\ \frac{u^2+v^2}{4\gamma e} - \frac{u}{2c} & \frac{1}{2c} - \frac{u}{2\gamma e} & -\frac{v}{2\gamma e} & \frac{1}{2\gamma e} \end{pmatrix}. \quad (2.19b)$$

Here, the average state $\mathbf{U}_{i+\frac{1}{2},j}$ is computed by the simple mean as following

$$\mathbf{U}_{i+\frac{1}{2},j} = \frac{1}{2} (\mathbf{U}_{i,j} + \mathbf{U}_{i+1,j}). \quad (2.20)$$

Then, we transform the conservation variables $\mathbf{U}_{k,j}$ to the local characteristic variables $\mathbf{V}_{k,j}$ by using the left eigenvectors L^F by

$$\mathbf{V}_{k,j} = L^F \mathbf{U}_{k,j}, \quad i-2 \leq k \leq i+3. \quad (2.21)$$

Furthermore, the local characteristic variables $\mathbf{V}_{i+\frac{1}{2},j}^\pm$ can be reconstructed. In general, the 5th order WENO reconstruction formulas of $\mathbf{V}_{i+\frac{1}{2},j}^-$ are given. And for this, the local characteristic variables $\mathbf{V}_{i-1,j}, \mathbf{V}_{i,j}, \mathbf{V}_{i+1,j}, \mathbf{V}_{i+2,j}$ and $\mathbf{V}_{i+3,j}$ are chosen, so we have

$$\mathbf{V}_{i+\frac{1}{2},j}^- = \omega_1 \mathbf{V}_{i+\frac{1}{2},j}^{(1)} + \omega_2 \mathbf{V}_{i+\frac{1}{2},j}^{(2)} + \omega_3 \mathbf{V}_{i+\frac{1}{2},j}^{(3)}, \quad (2.22)$$

where

$$\mathbf{V}_{i+\frac{1}{2},j}^{(1)} = -\frac{1}{6}\mathbf{V}_{i-1,j} + \frac{5}{6}\mathbf{V}_{i,j} + \frac{1}{3}\mathbf{V}_{i+1,j}, \quad (2.23a)$$

$$\mathbf{V}_{i+\frac{1}{2},j}^{(2)} = \frac{1}{3}\mathbf{V}_{i,j} + \frac{5}{6}\mathbf{V}_{i+1,j} - \frac{1}{6}\mathbf{V}_{i+2,j}, \quad (2.23b)$$

$$\mathbf{V}_{i+\frac{1}{2},j}^{(3)} = \frac{11}{6}\mathbf{V}_{i+1,j} - \frac{7}{6}\mathbf{V}_{i+2,j} + \frac{1}{3}\mathbf{V}_{i+3,j}, \quad (2.23c)$$

and

$$\omega_r = \frac{\sigma_r}{\sum_{r=1}^3 \sigma_r}, \quad \sigma_r = \frac{\theta_r}{(\epsilon + \beta_r)^2}, \quad (2.24)$$

and

$$\beta_1 = \frac{13}{12}(\mathbf{V}_{i-1,j} - 2\mathbf{V}_{i,j} + \mathbf{V}_{i+1,j})^2 + \frac{1}{4}(\mathbf{V}_{i-1,j} - 4\mathbf{V}_{i,j} + 3\mathbf{V}_{i+1,j})^2, \quad (2.25a)$$

$$\beta_2 = \frac{13}{12}(\mathbf{V}_{i,j} - 2\mathbf{V}_{i+1,j} + \mathbf{V}_{i+2,j})^2 + \frac{1}{4}(\mathbf{V}_{i+2,j} - \mathbf{V}_{i,j})^2, \quad (2.25b)$$

$$\beta_3 = \frac{13}{12}(\mathbf{V}_{i+1,j} - 2\mathbf{V}_{i+2,j} + \mathbf{V}_{i+3,j})^2 + \frac{1}{4}(3\mathbf{V}_{i+1,j} - 4\mathbf{V}_{i+2,j} + \mathbf{V}_{i+3,j})^2, \quad (2.25c)$$

and

$$\theta_1 = \frac{3}{10}, \quad \theta_2 = \frac{3}{5}, \quad \theta_3 = \frac{1}{10}. \quad (2.26)$$

Similarly with the reconstruction process of $\mathbf{V}_{i+\frac{1}{2},j}^-$, $\mathbf{V}_{i+\frac{1}{2},j}^+$ can also be reconstructed based on the local characteristic variables $\mathbf{V}_{i-2,j}$, $\mathbf{V}_{i-1,j}$, $\mathbf{V}_{i,j}$, $\mathbf{V}_{i+1,j}$ and $\mathbf{V}_{i+2,j}$.

Finally, we transform the local characteristic variables $\mathbf{V}_{i+\frac{1}{2},j}^\pm$ to the conservation variables $\mathbf{U}_{i+\frac{1}{2},j}^\pm$ by using the right eigenvectors R^F by

$$\mathbf{U}_{i+\frac{1}{2},j}^\pm = R^F \mathbf{V}_{i+\frac{1}{2},j}^\pm. \quad (2.27)$$

Thus we can get the flow variables $\rho_{i+\frac{1}{2},j}^\pm$, $u_{i+\frac{1}{2},j}^\pm$, $v_{i+\frac{1}{2},j}^\pm$ and $p_{i+\frac{1}{2},j}^\pm$.

The reconstruction process of the flow variables $\rho^\pm$, $u^\pm$, $v^\pm$ and $p^\pm$ in y -direction at point $(x_i, y_{j+\frac{1}{2}})$ are the same as in x -direction. Here, the local left eigenvectors and right

eigenvectors of the Jacobian matrix $\frac{\partial \mathbf{G}}{\partial \mathbf{U}}|_{x_i, y_{j+\frac{1}{2}}}$ are given by

$$R^G = \begin{pmatrix} 1 & 1 & 1 & 1 \\ u & 0 & 1 & u \\ v-c & v & v & v+c \\ \gamma e + \frac{u^2+v^2}{2} - vc & \frac{v^2-u^2}{2} & u + \frac{v^2-u^2}{2} & \gamma e + \frac{u^2+v^2}{2} + vc \end{pmatrix}, \quad (2.28a)$$

$$L^G = \begin{pmatrix} \frac{u^2+v^2}{4\gamma e} + \frac{v}{2c} & -\frac{u}{2\gamma e} & -\frac{1}{2c} - \frac{v}{2\gamma e} & \frac{1}{2\gamma e} \\ 1 - \frac{u^2+v^2}{2\gamma e} + \frac{u(u^2+v^2)}{2\gamma e} & \frac{u-u^2}{\gamma e} - 1 & \frac{v-uv}{\gamma e} & \frac{u-1}{\gamma e} \\ -\frac{u(u^2+v^2)}{2\gamma e} & 1 + \frac{u^2}{\gamma e} & \frac{uv}{\gamma e} & -\frac{u}{\gamma e} \\ \frac{u^2+v^2}{4\gamma e} - \frac{v}{2c} & -\frac{u}{2\gamma e} & \frac{1}{2c} - \frac{v}{2\gamma e} & \frac{1}{2\gamma e} \end{pmatrix}. \quad (2.28b)$$

Finally, the whole process of WENO-LBFS scheme is given as follows:

Step 1. Reconstruct the values $\rho^\pm, u^\pm, v^\pm$ and $p^\pm$ by WENO scheme:

Step 1.1 Transform the conservation variables $\mathbf{U}$ to the characteristic variables $\mathbf{V}$ based the local left characteristic vectors L^F and L^G by using Eqs. (2.19b), (2.28b) and (2.21);

Step 2.2 Reconstruct the characteristic variables $\mathbf{V}^\pm$ by using Eqs. (2.22)-(2.26);

Step 3.3 Transform the characteristic variables $\mathbf{V}^\pm$ to the conservation variables $\mathbf{U}^\pm$ based the local right characteristic vectors R^F and R^G by using Eqs. (2.19a), (2.28a) and (2.27);

Step 4.4 Transform the conservation variables $\mathbf{U}^\pm$ to the flow variables $\rho^\pm, u^\pm, v^\pm$ and $p^\pm$;

Step 2. Evaluate numerical fluxes $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$ based on non-free parameter D1Q4 model:

Step 2.1. Compute the density equilibrium distribution function f_α^R and lattice velocities e_α^R by using Eqs. (2.9)-(2.13);

Step 2.2. Evaluate the numerical fluxes $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$ by using Eqs. (2.15)-(2.18);

Step 3. Solve the spatial semi-discretization Eq. (2.4) by using the three-step Runge-Kutta scheme Eqs. (2.8) to update the conservation variables $\mathbf{U}$.

3 Numerical results

In this section, many numerical tests are performed to compare the results of WENO scheme based on the five different fluxes outlined in the previous section, including the

WENO-LBFS scheme, the WENO-LF scheme, the WENO-HLLC scheme, the WENO-MUSTA scheme and the WENO-FLIC scheme. The detailed numerical study is mainly performed for the one- and two-dimensional system cases. In numerical tests, the CFL numbers are taken as 0.45 and 0.475 for one- and two-dimensional states, respectively.

3.1 Accuracy test

First, we should have an accuracy test for the compressible flow problems with smooth solution to validate that the WENO-LBFS scheme should keep high-order accuracy and high efficiency, through comparing with conventional numerical fluxes (including LF, HLLC, MUSTA, FLIC, etc.).

Example 3.1. Consider the two dimensional Euler equations (2.1). The initial condition is set to be $\rho(x,y,0) = 1 + 0.2\sin(\pi(x+y))$, $u(x,y,0) = 0.7$, $v(x,y,0) = 0.3$, $p(x,y,0) = 1$, with a 2-periodic boundary condition. The final computational time is up to $t = 2$. The exact solution is $\rho(x,y,t) = 1 + 0.2\sin(\pi(x+y - (u+v)t))$, $u(x,y,t) = 0.7$, $v(x,y,t) = 0.3$, $p(x,y,t) = 1$.

The numerical errors, the orders of accuracy for the density and ratios of the numerical errors in comparison with the WENO-LBFS scheme are shown in Table 1. From this table, it can be discovered that the present WENO-LBFS scheme can achieve designed order of accuracy as well as others. All these schemes are fifth orders of convergence of L_1 and L_∞ errors except the WENO-FLIC scheme. The WENO-FLIC scheme can achieve the sixth order of convergence owing that it combines the Richtmyer flux as a high order flux and the FORCE flux as a low order monotone flux. However, it can be obviously found that the numerical error of the WENO-FLIC scheme is the smallest among all schemes for this case, which is followed by the WENO-HLLC scheme, while the error of the WENO-LF scheme is largest. Unfortunately, a limiter is introduced to determine the weights of the FORCE flux and the Richtmyer flux, which results that the WENO-FLIC scheme is unstable. With the grid refinement, the ratio of the numerical errors for the WENO-FLIC scheme compared with the WENO-LBFS schemes is range from 70% to 5%. The numerical error of the WENO-HLLC scheme is about half of that of the WENO-LBFS scheme, while that of the WENO-LF scheme is almost twice of the WENO-LBFS scheme. There is little difference between the WENO-LBFS scheme and the WENO-MUSTA scheme. That is to say, the WENO-LBFS scheme possess a good performance to solve the problem.

Furthermore, Fig. 1 shows that the relation curves between the numerical errors of density ρ and CPU time with different numerical fluxes. It can be seen that the relation curve of the WENO-LBFS scheme is closed to that of the WENO-MUSTA scheme. That is to say, the efficiency of these two schemes is close. Obviously, the efficiency of the WENO-LBFS scheme is much better than that of the WENO-LF scheme, and the WENO-HLLC scheme is better than the WENO-LBFS scheme. The WENO-FLIC scheme has the highest efficiency among all schemes for this problem with the smooth solution.

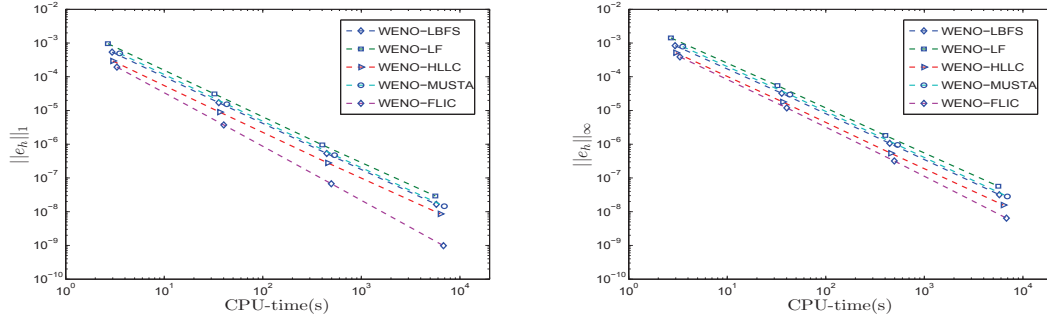


Figure 1: The relation curves of errors of density ρ and CPU time of different fluxes. Euler equations with initial condition: $\rho(x,y,0)=1+0.2\sin(\pi(x+y))$, $u(x,y,0)=0.7$, $v(x,y,0)=0.3$, $p(x,y,0)=1$, using $N=20,40,80,160$ equally spaced cells, $t=2$; Left: the relation curves of L_1 error and CPU time; Right: the relation curves of L_∞ error and CPU time.

Table 1: Euler equations with initial condition: $\rho(x,y,0)=1+0.2\sin(\pi(x+y))$, $u(x,y,0)=0.7$, $v(x,y,0)=0.3$, $p(x,y,0)=1$, using $N=10,20,40,80,160$, equally spaced cells with different fluxes, $t=2$, L_1 and L_∞ errors of density ρ .

N	Flux	L_1 error	L_1 order	Error ratio	L_∞ error	L_∞ order	Error ratio
10	WENO-LBFS	1.0243e-02		1.0000	1.3851e-02		1.0000
	WENO-LF	1.7779e-02		1.7358	2.5505e-02		1.8413
	WENO-HLLC	5.9357e-03		0.5795	9.4914e-03		0.6852
	WENO-MUSTA	9.4358e-03		0.9212	1.2799e-02		0.9240
	WENO-FLIC	7.0012e-03		0.6835	1.3020e-02		0.9400
20	WENO-LBFS	5.3830e-04	4.2500	1.0000	8.3720e-04	4.0483	1.0000
	WENO-LF	9.5643e-04	4.2164	1.7767	1.4142e-03	4.1727	1.6892
	WENO-HLLC	2.9383e-04	4.3364	0.5458	5.1566e-04	4.2021	0.6159
	WENO-MUSTA	4.9317e-04	4.2580	0.9161	7.8169e-04	4.0333	0.9337
	WENO-FLIC	1.9294e-04	5.1813	0.3584	3.8761e-04	5.0700	0.4630
40	WENO-LBFS	1.7060e-05	4.9797	1.0000	3.2370e-05	4.6929	1.0000
	WENO-LF	3.0983e-05	4.9481	1.8162	5.4655e-05	4.6935	1.6885
	WENO-HLLC	8.9671e-06	5.0342	0.5256	1.7917e-05	4.8470	0.5535
	WENO-MUSTA	1.5318e-05	5.0088	0.8979	2.9489e-05	4.7283	0.9110
	WENO-FLIC	3.7306e-06	5.6927	0.2187	1.1918e-05	5.0233	0.3682
80	WENO-LBFS	5.3237e-07	5.0020	1.0000	1.0617e-06	4.9302	1.0000
	WENO-LF	9.5172e-07	5.0248	1.7877	1.8213e-06	4.9073	1.7155
	WENO-HLLC	2.7836e-07	5.0096	0.5229	5.4166e-07	5.0478	0.5102
	WENO-MUSTA	4.7280e-07	5.0178	0.8881	9.5116e-07	4.9544	0.8959
	WENO-FLIC	6.7351e-08	5.7915	0.1265	3.1885e-07	5.2242	0.3003
160	WENO-LBFS	1.6455e-08	5.0158	1.0000	3.1908e-08	5.0563	1.0000
	WENO-LF	2.9223e-08	5.0254	1.7759	5.6760e-08	5.0040	1.7789
	WENO-HLLC	8.6006e-09	5.0164	0.5227	1.5751e-08	5.1039	0.4936
	WENO-MUSTA	1.4550e-08	5.0222	0.8842	2.8298e-08	5.0709	0.8868
	WENO-FLIC	9.8354e-10	6.0976	0.0598	6.4415e-09	5.6294	0.2019

3.2 Test 1D cases with shocks

As we all know that the compressible flow problems usually arise shock waves, which demand that the numerical algorithms are robust. This robustness often performs avoiding numerical oscillations and high resolution for shock-capturing. Here, three classical shock problems are presented to test the performance of the WENO-LBFS scheme. Meanwhile, the WENO-LBFS scheme is compared with WENO scheme based on conventional numerical fluxes.

Example 3.2. Lax problem

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} = 0, \quad (3.1)$$

where

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho E \end{pmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ (\rho E + p)u \end{pmatrix}, \quad (3.2)$$

with the initial condition:

$$(\rho, u, p) = \begin{cases} (0.445, 0.698, 3.528) & \text{for } x \leq 0, \\ (0.5, 0, 0.571) & \text{for } x > 0. \end{cases} \quad (3.3)$$

The computational domain is $x \in [-5, 5]$ covered with $N = 200$ grid points. The compact boundary condition is imposed for the left and right boundaries. The simulation time is up to 1.3.

The classical elementary waves in the solution of the Lax problem include rarefaction wave, shock wave and contact discontinuity. In Fig. 2, the densities ρ at time $t = 1.3$ are plotted against the exact solution, where the comparisons of LBFS and conventional numerical fluxes are described clearly. These figures are zoomed at the region $x \in [1, 4]$, $\rho \in [0.2, 1.4]$ which contains shock wave and contact discontinuity to facilitate comparative analysis. From Fig. 2, it is easy to know that the results computed by the WENO-LBFS scheme are better than that computed by the WENO-LF scheme. The WENO-FLIC scheme has the highest resolution among all schemes, followed closely by the WENO-MUSTA scheme. There is little difference between the WENO-MUSTA scheme and the WENO-HLLC scheme.

Example 3.3 (Shu-Osher problem). The initial condition is

$$(\rho, u, p) = \begin{cases} (3.857143, 2.629369, 10.333333) & \text{for } x \leq -4, \\ (1 + \varepsilon \sin(0.5x), 0, 1) & \text{for } x > -4, \end{cases} \quad (3.4)$$

where ε is set to be 0.2. The computational domain is $[-5, 5]$ covered with $N = 300$ grid points, and left is compact boundary and right is inflow boundary.

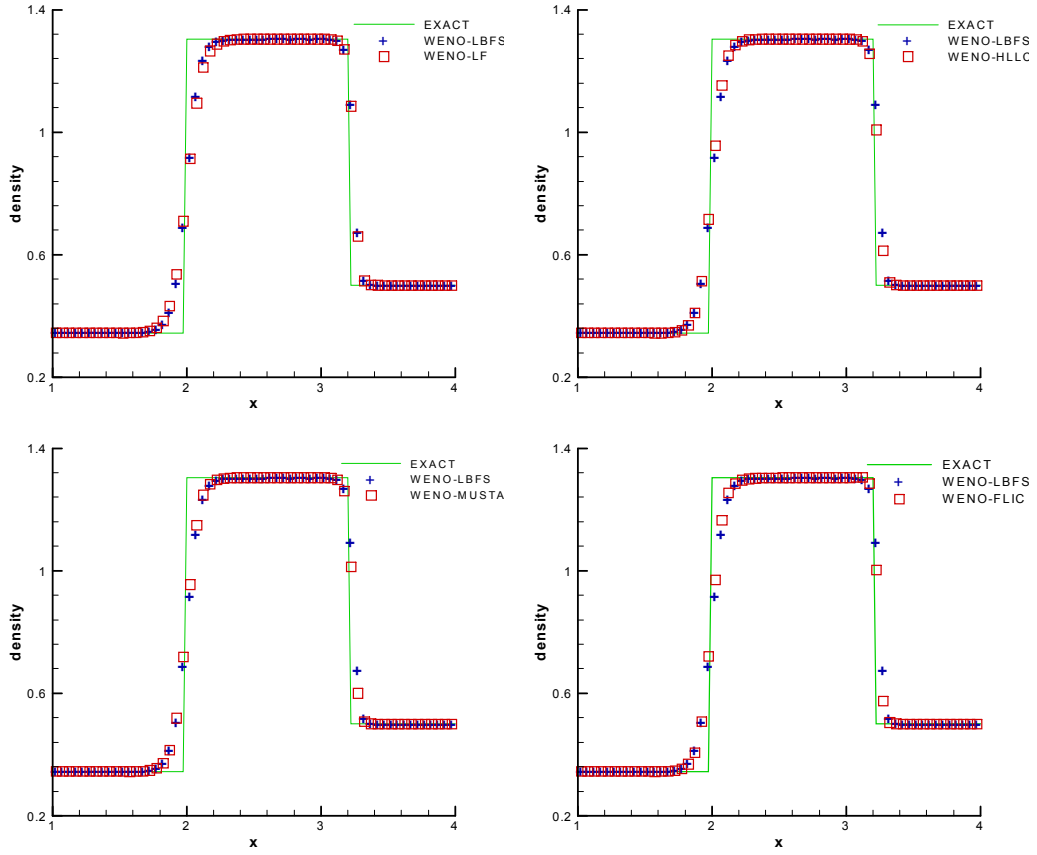


Figure 2: Lax problem. $t=1.3$. WENO with different fluxes, 200 cells. Density. Solid line: the exact solution; Plus symbols: the results computed by the WENO-LBFS scheme; Hollow squares: the results computed by the WENO-LF scheme, the WENO-HLLC scheme, the WENO-MUSTA scheme and the WENO-FLIC scheme.

The classical elementary waves in the solution of the Shu-Osher problem include high-frequency wave and shock wave. Fig. 3 depicts the density ρ at time $t=0.2$ against the reference solution computed by the WENO-LF scheme using 2000 grids. These figures are zoomed at the region $x \in [0.5, 2.5]$, $\rho \in [2.5, 5.0]$ which contains high-frequency wave to facilitate comparative analysis. The comparisons of LBFS and conventional numerical fluxes are described clearly in Fig. 3. From these figures, it can be found that the WENO-LBFS scheme has better performance to capture the high-frequency wave than the WENO-LF scheme, but there is no significant difference between the WENO-LBFS scheme and conventional WENO schemes including the WENO-HLLC scheme and the WENO-MUSTA scheme. The WENO-FLIC scheme is better than the WENO-LBFS scheme for the problem with the smooth solution.

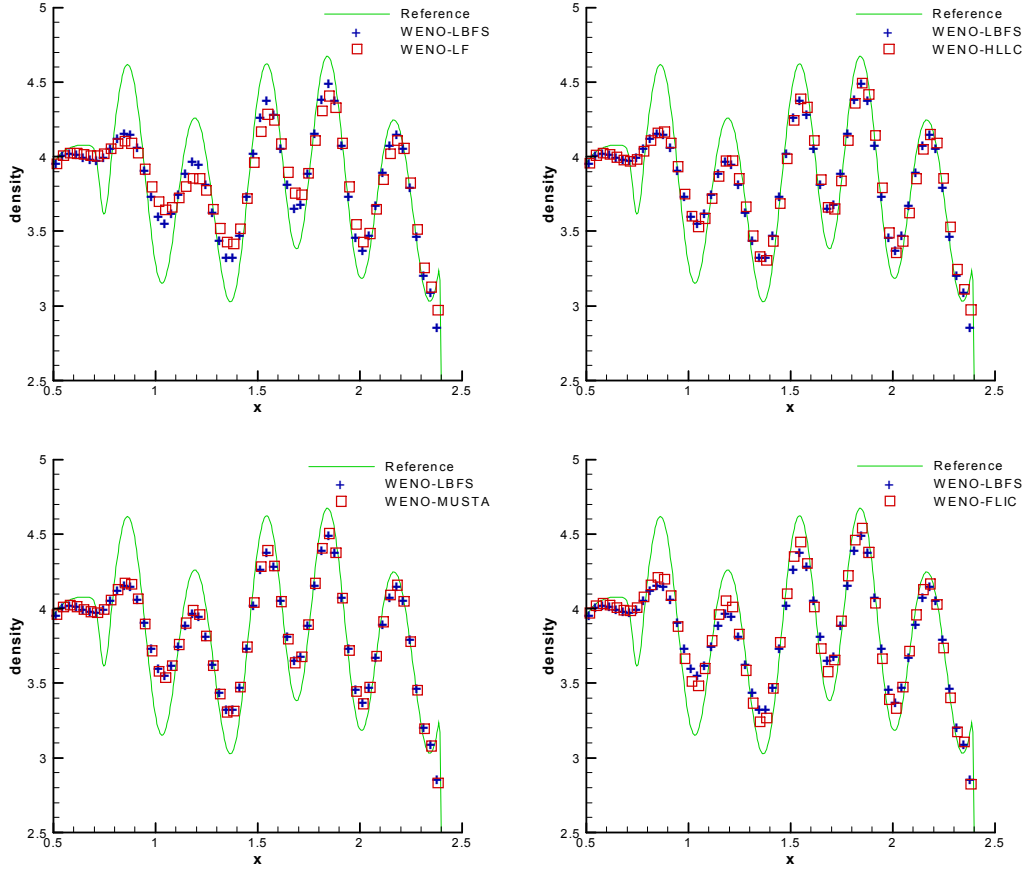


Figure 3: Shu-Osher problem. $t=1.8$. WENO with different fluxes, 300 cells. Density. Solid line: the "exact" reference solution; Plus symbols: the results computed by the WENO-LBFS scheme; Hollow squares: the results computed by the WENO-LF scheme, the WENO-HLLC scheme, the WENO-MUSTA scheme and the WENO-FLIC scheme.

Example 3.4. Woodward-Colella blast wave problem [37]. The initial condition is

$$(\rho, u, p) = \begin{cases} (1, 0, 1000) & \text{for } 0 \leq x < 0.1, \\ (1, 0, 0.01) & \text{for } 0.1 \leq x < 0.9, \\ (1, 0, 100) & \text{for } x \geq 0.9. \end{cases} \quad (3.5)$$

The computational domain is $x \in [0, 1]$ covered with $N=300$ grid points. Reflective boundary condition is used for the left and right boundaries. The final simulation time is up to 0.038.

The solution of Woodward problem evolves complex wave structures yielded by the interaction of two shock waves. In Fig. 4, the densities ρ at time $t = 0.038$ are plotted against the reference solution computed by the WENO-LF scheme using 2000 grids.

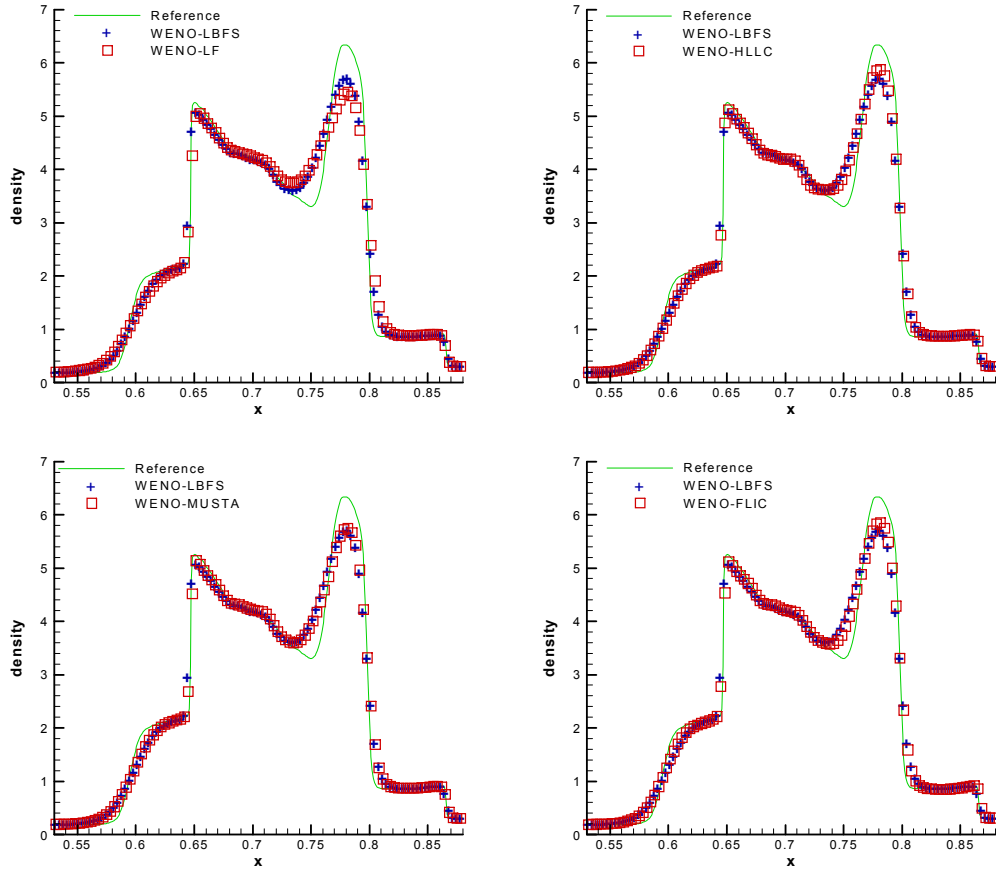


Figure 4: Woodward problem. $t = 0.038$. WENO with different fluxes, 300 cells. Density. Solid line: the "exact" reference solution; Plus symbols: the results computed by the WENO-LBFS scheme; Hollow squares: the results computed by the WENO-LF scheme, the WENO-HLLC scheme, the WENO-MUSTA scheme and the WENO-FLIC scheme.

These figures are zoomed at the region $x \in [0.53, 0.88]$, $\rho \in [0, 7]$ which contains complex wave structures to facilitate comparative analysis. The comparisons of LBFS and conventional numerical fluxes are described clearly in Fig. 4. Obviously, for the left contact discontinuity, the resolution of the WENO-LBFS scheme is far better than that of the WENO-LF scheme and slightly better than that of the WENO-MUSTA scheme and the WENO-FLIC scheme, while the resolution of the WENO-HLLC scheme is better than that of the WENO-LBFS scheme. For the right shock wave, the result computed by the WENO-LBFS scheme is better than that computed by the WENO-LF scheme. There is barely a difference between the WENO-LBFS scheme and the WENO-MUSTA scheme. The result computed by the WENO-FLIC scheme is best, followed closely by that of the WENO-HLLC scheme.

3.3 Test 2D cases with shocks

In the previous section, 1D cases with shocks are tested. Based on the numerical results, it is known that the WENO-LBFS scheme has good performance to capture some complex wave structures (including rarefaction waves, shock waves, contact discontinuities and high-frequency waves, etc.). Next, 2D cases with shocks, the wave structures of which are more complex, are used to test the WENO-LBFS scheme. The results computed by the WENO-LBFS scheme are compared with the results computed by conventional fluxes.

Example 3.5. Double Mach reflection [34]. The computational domain is $[0,4] \times [0,1]$. Initially a right-moving Mach 10 shock is positioned at $x = \frac{1}{6}, y = 0$ and makes a 60° angle with x -axis. For the bottom boundary, the exact post-shock condition is imposed for the range $0 \leq x \leq \frac{1}{6}$, and the reflecting wall lies the rest. For the top boundary, the flow values are set to describe the exact motion of a Mach 10 shock. For the left and right boundaries, inflow boundary and outflow boundary are adopted, respectively. The computational time is up to $t = 0.2$.

This problem is often provided as a test case for high-resolution schemes. Fig. 5 shows the density contours with grid size $h = \frac{1}{480}$ for WENO scheme based on different numerical fluxes. Especially, the solution obtained by the WENO-LF scheme with grid size $h = \frac{1}{960}$ as the reference solution is given. The figures are zoomed at the region $[2,2.875] \times [0,1]$ which contains complex wave structures to facilitate comparative analysis, and the density is plotted by 30 equally spaced contour lines from $\rho = 1.5$ to $\rho = 22.8$. Obviously, the WENO-LBFS scheme and the WENO-MUSTA scheme have a significantly better performance to capture the complex wave structures than the WENO-LF scheme. For the WENO-HLLC scheme, it should be noted that there are more vortices at the corner of reflection shock wave, which is not what we expect according to the reference solution. For the WENO-FLIC scheme, it can capture more complex wave structures than other schemes, while these vortices are greatly smoothed.

Example 3.6 (Implosion problem [23]). The initial condition is

$$(\rho, u, v, p) = \begin{cases} (1, 0, 0, 1), & |x| - 0.15 \leq y \leq -|x| + 0.15, \\ (0.125, 0, 0, 0.14), & \text{otherwise.} \end{cases} \quad (3.6)$$

The computational domain is $[-0.3, 0.3] \times [-0.3, 0.3]$. Reflective boundary condition is imposed for the left, right and bottom boundaries. For the top boundary, outflow boundary is applied. The final simulation time is up to 0.8.

Implosion problem is used to illustrate the ability of the WENO-LBFS scheme for two-dimensional problems. Fig. 6 shows the pressure and Mach number contours with the uniform grid size of 400×400 for WENO scheme based on different numerical fluxes. Especially, the solution obtained by the WENO-LF scheme with the uniform grid size of 800×800 as the reference solution is given. The pressure is plotted by 35 equally spaced

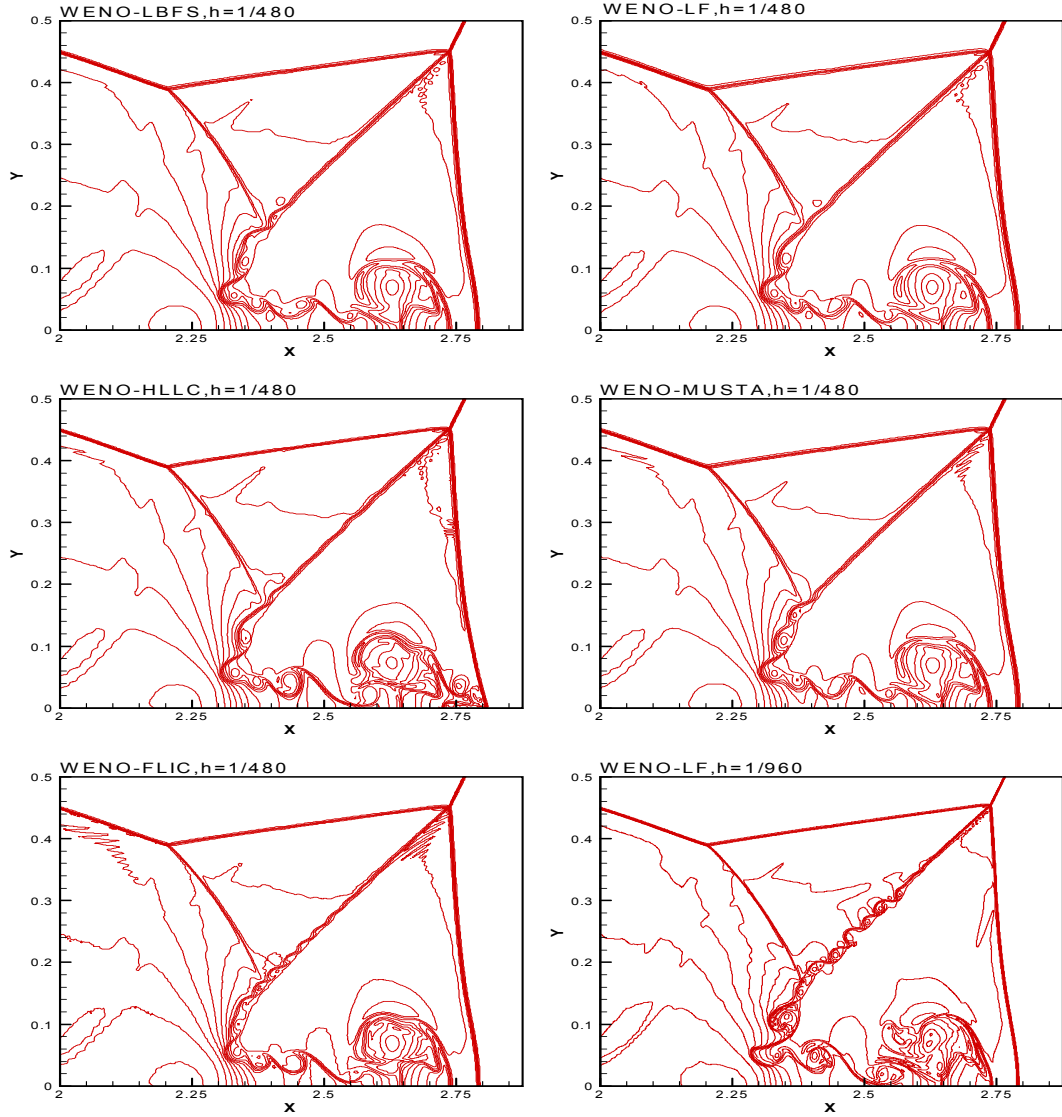


Figure 5: Double Mach reflection problem. $t=0.2$. Density ρ : 30 equally spaced contour lines from $\rho=1.5$ to $\rho=22.8$. Left: the results with $h=\frac{1}{480}$ computed by the WENO-LBFS scheme, the WENO-HLLC scheme and the WENO-FLIC scheme; Right: the results with $h=\frac{1}{480}$ computed by the WENO-LF scheme and the WENO-MUSTA scheme, and the result with $h=\frac{1}{960}$ computed by the WENO-LF scheme.

contour lines from $p=0.54$ to $p=1.19$, and the Mach number is depicted by same contour lines from $Mach=0.01$ to $Mach=0.2$ to facilitate comparative analysis. Obviously, the WENO-LBFS scheme has a significantly better performance to capture the complex wave structures than the WENO-LF scheme, and there is no significant difference between the WENO-LBFS scheme and the WENO-MUSTA scheme. The WENO-HLLC scheme

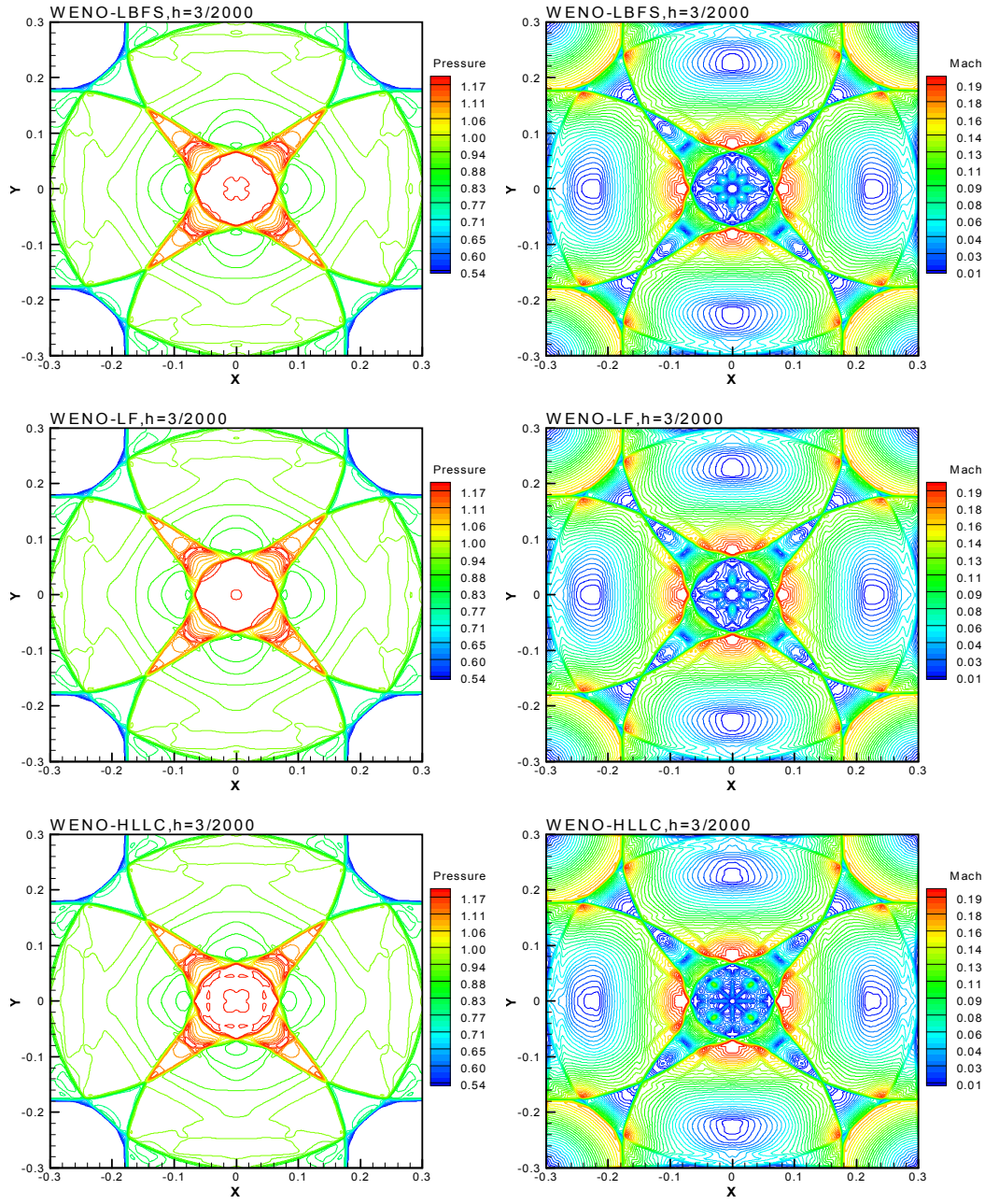


Figure 6: Implosion problem. $t=0.8$. Pressure: 35 equally spaced contour lines from $p=0.54$ to $p=1.19$. Mach number: 35 equally spaced contour lines from $Mach=0.01$ to $Mach=0.2$. Left: the pressure computed by the WENO-LBFS scheme, the WENO-LF scheme and the WENO-HLLC scheme with grid size of 400×400 ; right: the mach number computed by these three schemes with grid size of 400×400 .

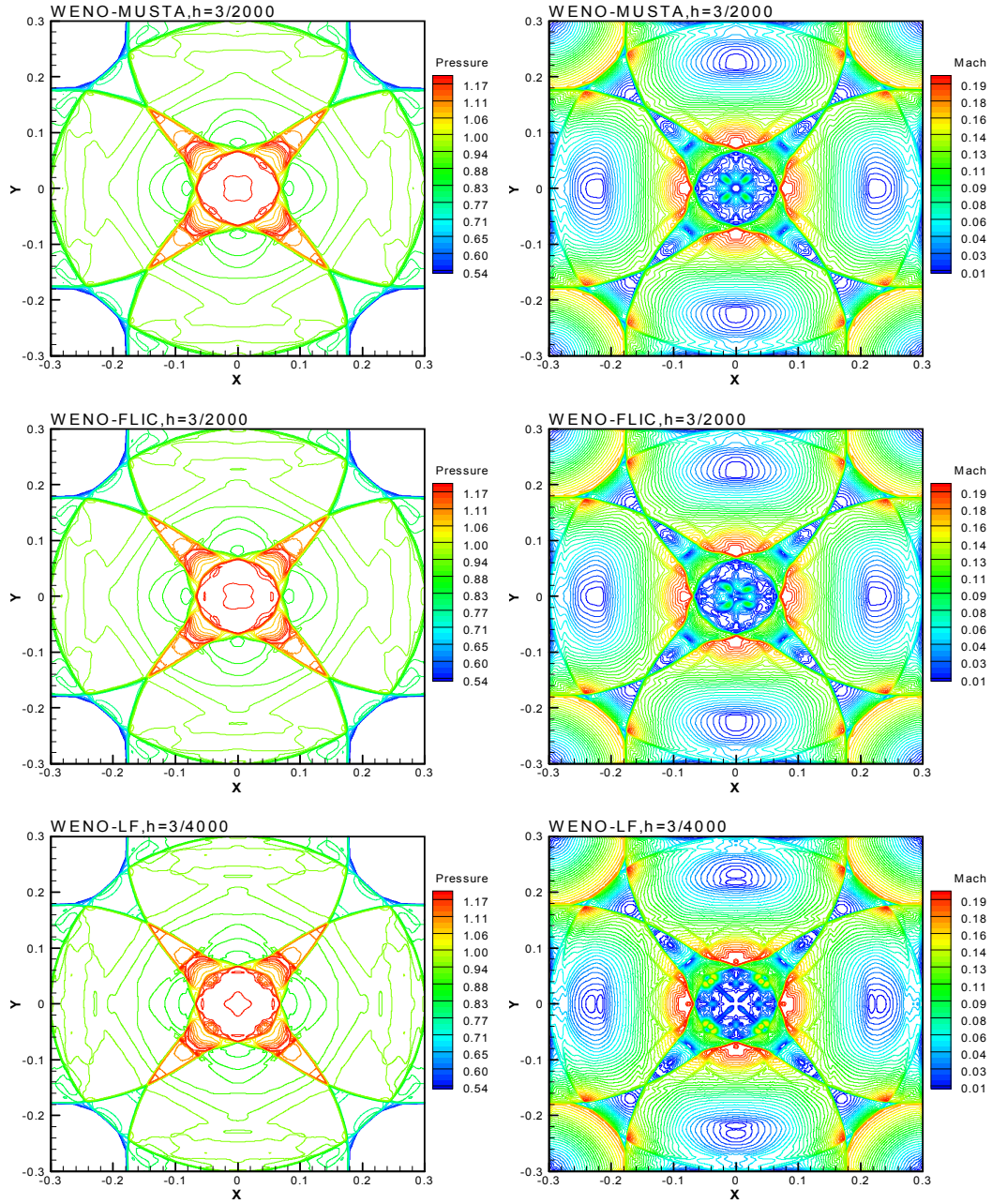


Figure 7: Implosion problem. $t=0.8$. Pressure: 35 equally spaced contour lines from $p=0.54$ to $p=1.19$. Mach number: 35 equally spaced contour lines from $Mach=0.01$ to $Mach=0.2$. Left: the pressure computed by the WENO-MUSTA scheme and the WENO-FLIC scheme with grid size of 400×400 , and by the WENO-LF scheme with grid size of 800×800 ; right: the Mach number computed by the WENO-MUSTA scheme and the WENO-FLIC scheme with grid size of 400×400 , and by the WENO-LF scheme with grid size of 800×80 .

is slightly better than the WENO-LBFS scheme. For the WENO-FLIC scheme, it should be noted that it is not our expectation according to the reference solution that the wave structures at the center are not symmetrical.

4 Conclusions

In this paper, the WENO-LBFS scheme, which is used for simulation of one- and two-dimensional compressible flows, is constructed. The present scheme combines the advantages of WENO scheme and lattice Boltzmann flux solver (LBFS). WENO scheme is a high-order and stable scheme. For capturing shock waves, WENO scheme not only has high resolution, but also avoids the numerical oscillations. LBFS is based on the local solution of lattice Boltzmann equation (LBE), and it is more physically robust in comparison with conventional numerical fluxes that are based on mathematical interpolations. Based on the above, the WENO-LBFS scheme is destined to be a high-order accuracy and high-resolution numerical scheme to capture shock waves,

A series of comparative studies are conducted to evaluate the advantage of the WENO-LBFS scheme over other WENO scheme-based flux solvers. One-dimensional simulation with smooth solution indicates that the WENO-LBFS scheme can achieve fifth order accuracy. What's more, the WENO-LF scheme costs the least CPU time among all schemes, and there is barely a difference between the WENO-LBFS scheme and the WENO-LF scheme, but the numerical error and resolution of the WENO-LBFS scheme are far better than that of the WENO-LF scheme.

Furthermore, extensive one- and two- dimensional simulations with shock waves indicate that the results computed by the WENO-LBFS scheme are far superior to that of the WENO-LF scheme and are nor inferior to that of WENO scheme with other fluxes, including the WENO-MUSTA scheme, the WENO-HLLC scheme and the WENO-FLIC scheme. Most importantly, the computational efficiency of the WENO-LBFS scheme is the highest and its programming is the simplest except for the WENO-LF scheme.

Above all, the WENO-LBFS scheme is a high-order accuracy, high-resolution, high-efficient and robust scheme. The WENO-LBFS scheme is a good choice for simulating compressible flows with shock waves and discontinuities when all factors such as the cost of CPU time, numerical errors and resolution near discontinuities and shock waves in the solution are considered. It should be noted that, in this paper, the evaluation of the performance of the WENO-LBFS scheme is only restricted to compressible flows. The WENO-LBFS scheme for incompressible flows and multi-phase flows will be discussed in the future.

Acknowledgments

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References

- [1] A. Harten, B. Engquist, S. Osher and S. R.Chakravarthy, Uniformly high order accurate essentially non-oscillatory schemes, iii, *J. Comput. Phys.*, 131 (1997), 3–47.
- [2] A. Harten, P. D. Lax and B. van-Leer, On upstream differencing and godunov-type schemes for hyperbolic conservation laws, *Society for Industrial and Applied Mathematics*, 25(1) (1983), 35–61.
- [3] B. Engquist and S. Osher, One-sided difference approximations for nonlinear conservation laws, *Math. Comput.*, 36(154) (1981), 321–351.
- [4] C. Q. Hu and C. W. Shu, Weighted essentially non-oscillatory schemes on triangular meshes, *J. Comput. Phys.*, 150 (1999), 97–127.
- [5] C. Shu, Y. Wang, C. J. Teo and J. Wu, Development of lattice boltzmann flux solver for simulation of incompressible flows, *Adv. Appl. Math. Mech.*, 6(4) (2014), 436–460.
- [6] C. Shu, Y. Wang, L. M. Yang and J. Wu, Lattice boltzmann flux solver:an efficient approach for numerical simulation of fluid flows, *Transactions of Nanjing University of Aeronautics and Astronautics*, 31(1) (2014), 1–15.
- [7] C. W. Shu, Essentially non-oscillatory and weighted essentially non-oscillatory schemes for hyperbolic conservation laws, *Institute Comput. Appl. Sci. Report*, 1997 (1997), 97–65.
- [8] C. W. Shu and S. Osher, Efficient implementation of essentially non-oscillatory shock-capturing schemes, *J. Comput. Phys.*, 77(2) (1988), 439–471.
- [9] C. Z. Ji, C. Shu and N. Zhao, A lattice boltzmann method-based flux solver and its application to solve shock tube problem, *Modern Phys. Lett. B*, 23(3) (2009), 313–316.
- [10] D. Balsara and C. W. Shu, Monotonicity preserving weighted essentially non high-oscillatory schemes with increasingly high order accuracy, *J. Comput. Phys.*, 160(2) (2000), 405–452.
- [11] D. L. Zhang, *A Course in Computational Fluid Dynamics*, Higher Education Press, 2010.
- [12] E. F. Toro, Multi-stage predictor-corrector fluxes for hyperbolic equations, Preprint NI03037-NPA, Issac Newton Institute for Mathematical Sciences, University of Cambridge, UK, 2003.
- [13] E. F. Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics: A Practical Introduction*, Springer-Verlag Berlin Heidelberg, 2009.
- [14] E. F. Toro, M. Spruce and W. Speares, Restoration of the contact surface in the harten-lax-van leer riemann solve, *Shock Waves*, 4(1) (1994), 25–34.
- [15] G. S. Jiang and C. W. Shu, Efficient implementation of weighted ENO schemes, *J. Comput. Phys.*, 126 (1996), 202–228.
- [16] J. Shi, C. Q. Hu and C. W. Shu, A technique of treating negative weights in WENO schemes, *J. Comput. Phys.*, 175 (2002), 108–127.
- [17] J. VonNeumann and R. D. Richtmyer, A method for the numerical calculation of hydrodynamic shocks, *J. Appl. Phys.*, 21(3) (1950), 232–237.
- [18] J. X. Qiu, B. C. Khoo and C. W. Shu, A numerical study for the performance of the runge-kutta discontinuous galerkin method based on different numerical fluxes, *J. Comput. Phys.*, 212(2) (2006), 540–565.
- [19] K. Qu, C. Shu and Y. T. Chew, Alternative method to construct equilibrium distribution functions in lattice-boltzmann method simulation of inviscid compressible flows at high mach number, *Phys. Rev. E*, 75(3) (2007), 036706.

- [20] K. Qu, C. Shu and Y. T. Chew, Simulation of shock-wave propagation with finite volume lattice boltzmann method, *Int. J. Modern Phys. C*, 18(4) (2007), 447–454.
- [21] K. Xu, A gas-kinetic BGK scheme for the navier-stocks equations and its connection with artificial dissipation and godunov method, *J. Comput. Phys.*, 171 (2001), 289–335.
- [22] L. B. Van, Towards to the ultimate conservation difference schemes: a second-order sequel to godunov method, *J. Comput. Phys.*, 135(2) (1997), 229–248.
- [23] L. M. Yang, C. Shu and J. Wu, Development and comparative studies of three non-free parameter lattice boltzmann models for simulation of compressible flows, *Adv. Appl. Math. Mech.*, 4(4) (2012), 454–472.
- [24] L. M. Yang, C. Shu and J. Wu, A moment conservation-based non-free parameter compressible lattice boltzmann model and its application for flux evaluation at cell interface, *Comput. Fluids*, 79 (2013), 190–199.
- [25] L. M. Yang, C. Shu and J. Wu, A simple distribution function-based gas-kinetic scheme for simulation of viscous incompressible and compressible flows, *J. Comput. Phys.*, 274 (2014), 611–632.
- [26] L. M. Yang, C. Shu and J. Wu, Extension of lattice boltzmann flux solver for simulation of 3d viscous compressible flows, *Comput. Math. Appl.*, (2016).
- [27] L. M. Yang, C. Shu and J. Wu, A hybrid lattice boltzmann flux solver for simulation of viscous compressible flows, *Adv. Appl. Math. Mech.*, 8(6) (2016), 887–910.
- [28] L. M. Yang, C. Shu, J. Wu, N. Zhao and Z. L. Lu, Circular function-based gas-kinetic scheme for simulation of inviscid compressible flows, *J. Comput. Phys.*, 255 (2013), 540–557.
- [29] L. M. Yang, C. Shu, Y. Wang and Y. Sun, Development of discrete gas kinetic scheme for simulation of 3d viscous incompressible and compressible flows, *J. Comput. Phys.*, 319 (2016), 129–144.
- [30] M. Käser and M. Dumbser, An arbitrary high order discontinuous galerkin method for elastic waves on unstructured meshes i: the two-dimensional isotropic case with external source terms, *Geophys. J. Int.*, 166(2) (2006), 855–877.
- [31] O. Friedrich, Weighted essentially non-oscillatory schemes for the interpolation of mean values on unstructured grids, *J. Comput. Phys.*, 144(1) (1998), 194–212.
- [32] P. Colella and P. R. Woodward, The piecewise parabolic method (ppm) for gas-dynamical simulations, *J. Comput. Phys.*, 54(1) (1984), 174–201.
- [33] P. Lax and B. Wendroff, Systems of conservation laws, *Commun. Pure Appl. Math.*, 13(2) (1960), 217–237.
- [34] P. Woodward and P. Colella, The numerical simulation of two-dimensional fluid flow with strong shocks, *J. Comput. Phys.*, 54(1) (1984), 115–173.
- [35] S. K. Godunov, A difference method for numerical computation of discontinuous solution of hydrodynamic equations, *Matematicheskii Sbornik*, 89(3) (1959), 271–306.
- [36] S. Osher and F. Solomon, Upwind difference schemes for hyperbolic systems of conservation laws, *Math. Comput.*, 38(158) (1982), 339–374.
- [37] V. A. Titarev and E. F. Toro, Finite-volume weno schemes for three-dimensional conservation laws, *J. Comput. Phys.*, 201(1) (2004), 238–260.
- [38] V. A. Titarev and E. F. Toro, Weno schemes based on upwind and centred TVD fluxes, *Comput. Fluids*, 34(6) (2005), 705–720.
- [39] W. H. Reed and T. R. Hill, Triangular mesh methods for the neutron transport equation, Los Alamos Report LA-UR-73-479, 1973.
- [40] X. D. Liu, S. Osher and T. Chan, Weighted essentially non-oscillatory schemes, *J. Comput. Phys.*, 115(1) (1994), 200–212.

- [41] Y. Gan, A. Xu, G. Zhang and Y. Li, Lattice boltzmann study on kelvin-helmholtz instability: the roles of velocity and density gradients, *Phys. Rev. E*, 83(5) (2011), 056704.
- [42] Y. Sun, C. Shu, C. J. Teo, Y. Wang and L. M. Yang, Explicit formulations of gas-kinetic scheme for simulation of incompressible and compressible viscous flows, *J. Comput. Phys.*, 300 (2015), 492–519.
- [43] Y. Sun, C. Shu, L. M. Yang and C. J. Teo, A switch function-based gas-kinetic scheme for simulation of inviscid and viscous compressible flows, *Adv. Appl. Math. Mech.*, 8(5) (2016), 703–721.
- [44] Y. Wang, C. Shu, C. J. Teo and L. M. Yang, An efficient immersed boundary-lattice boltzmann flux solver for simulation of 3d incompressible flows with complex geometry, *Comput. Fluid*, 124 (2015), 54–66.
- [45] Y. Wang, Y. L. He, T. S. Zhao, G. H. Tang and W. Q. Tao, Implicit-explicit finite-difference lattice boltzmann method for compressible flows, *Int. J. Modern Phys. C*, 18(12) (2007), 1961–1983.